MEDIUM

PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

02

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 PROBLEM-SOLVING AND DATA ANALYSIS

Station 1	Station 2	Station 3	Station 4	Station 5
$3.699	$3.609	$3.729	$3.679	$3.729

In the table above, Melissa recorded the price of one gallon of regular gas from five different local gas stations on the same day. What is the median of the gas prices Melissa recorded?

A. $3.679

B. $3.689

C. $3.699

D. $3.729

Q2

PROBLEM-SOLVING AND DATA ANALYSIS

A distance of 354 furlongs is equivalent to how many feet?

$1 \text{ furlong} = 220 \text{ yards}$ and $1 \text{ yard} = 3 \text{ feet}$

A. 306

B. 402

C. 25,960

D. 233,640

Q3

PROBLEM-SOLVING AND DATA ANALYSIS

A study was done on the weights of different types of fish in a pond. A random sample of fish were caught and marked in order to ensure that none were weighed more than once. The sample contained 150 largemouth bass, of which 30% weighed more than 2 pounds. Which of the following conclusions is best supported by the sample data?

A. The majority of all fish in the pond weigh less than 2 pounds.

B. The average weight of all fish in the pond is approximately 2 pounds.

C. Approximately 30% of all fish in the pond weigh more than 2 pounds.

D. Approximately 30% of all largemouth bass in the pond weigh more than 2 pounds.

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

A survey was conducted using a sample of history professors selected at random from the California State Universities. The professors surveyed were asked to name the publishers of their current texts. What is the largest population to which the results of the survey can be generalized?

A. All professors in the United States

B. All history professors in the United States

C. All history professors at all California State Universities

D. All professors at all California State Universities

Q5

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

There are 450 tiles in a box. Of these tiles, 6% are black. How many black tiles are in the box?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The table below shows the number of state parks in a certain state that contain camping facilities and bicycle paths.

	Has bicycle paths	Does not have bicycle paths
Has camping facilities	20	5
Does not have camping facilities	8	4

If one of these state parks is selected at random, what is the probability that it has camping facilities but does not have bicycle paths?

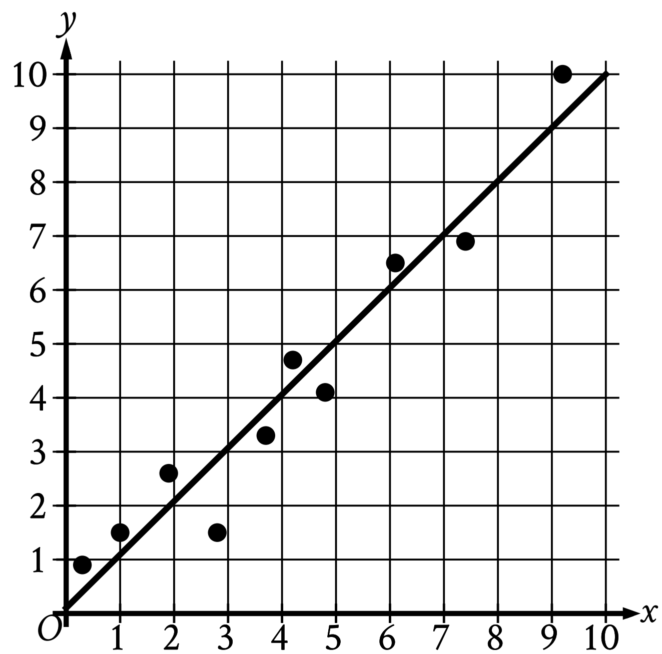
A. $\frac{5}{37}$

B. $\frac{5}{25}$

C. $\frac{8}{28}$

D. $\frac{5}{9}$

The scatterplot shows the relationship between two variables, x and y. A line of best fit for the data is also shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
    - The line of best fit slants up from left to right.
    - 6 points are above the line of best fit.
    - 4 points are below the line of best fit.
    - The line of best fit passes through the following approximate coordinates:
        - (0 comma 0.1)
        - (5 comma 5.1)

For how many of the 10 data points is the actual y-value greater than the y-value predicted by the line of best fit?

A. 3

B. 4

C. 6

D. 7

Q8

PROBLEM-SOLVING AND DATA ANALYSIS

The results of two independent surveys are shown in the table below.

Men's Height

Group	Sample size	Mean (centimeters)	Standard deviation (centimeters)
A	2,500	186	12.5
B	2,500	186	19.1

Which statement is true based on the table?

A. The Group A data set was identical to the Group B data set.

B. Group B contained the tallest participant.

C. The heights of the men in Group B had a larger spread than the heights of the men in Group A.

D. The median height of Group B is larger than the median height of Group A.

Q9

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

A distance of 112 furlongs is equivalent to how many feet?

$1 \text{ furlong} = 220 \text{ yards}$ and $1 \text{ yard} = 3 \text{ feet}$

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

A random sample of 400 town voters were asked if they plan to vote for Candidate A or Candidate B for mayor. The results were sorted by gender and are shown in the table below.

	Plan to vote for Candidate A	Plan to vote for Candidate B
Female	202	20
Male	34	144

The town has a total of 6,000 voters. Based on the table, what is the best estimate of the number of voters who plan to vote for Candidate A?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.